

Inference by Enum:

Example Knowledgebase 1: $\neg P(1,1) \vee B(1,2) \vee \neg B(1,1)$

?2,1	?2,2		
$\neg P \vee B$	B	?1,3	

Alpha: $P(2,2)$ {PIT at 2,2}

Models

Here we can say $\{ B(1,2) \Rightarrow [P(2,2) \vee P(1,3)] \}$

$\{ \neg B(1,1) \Rightarrow [\neg P(2,1) \wedge \neg P(1,2)] \}$

SO Knowledgebase Condition:- $[P(2,2) \vee P(1,3)] \wedge \neg P(2,1)$

P2,1	P2,2	P1,3	KB $[P(2,2) \vee P(1,3)] \wedge \neg P(2,1)$	Alpha $P(2,2)$
False	False	False	True	F
False	False	True	True	F
False	True	False	True	True
False	True	True	True	True
True	False	False	False	F
True	False	True	False	F
True	True	False	False	True
True	True	True	False	True

Here KB is not giving 100% guarantee that there is a pit in (2,2) .in some cases KB true but alpha false. So KB does not entail alpha.

Practice for same knowledgebase with query: $[P(2,2) \vee P(1,3)]$, $[P(2,2) \wedge P(1,3)]$

Inference by Enum:

Example Knowledgebase 1: $\neg P(1,1) \neg P(1,2) B(1,1) B(1,2)$

?2,1	?2,2		
B $\neg$ P	B $\neg$ P	?1,3	

Alpha: $P(2,1)$ {PIT at 2,1}

Models

Here we can say $\{ B(1,1) \text{ and } \neg P(1,2) \Rightarrow P(2,1) [100\%] \}$

$\{ \neg P(1,1) \text{ and } B(1,2) \Rightarrow [P(2,2) \text{ or } P(1,3)] \}$

SO Knowledgebase Condition:- $[P(2,2) \text{ or } P(1,3)] \text{ and } P(2,1)$

P2,1	P2,2	P1,3	KB [$P(2,2)$ or $P(1,3)$] and $P(2,1)$]	Alpha $P(2,2)$
False	False	False	False	False
False	False	True	False	False
False	True	False	False	False
False	True	True	False	False
True	False	False	True	True
True	False	True	True	True
True	True	False	True	True
True	True	True	True	True

Here KB is giving 100% guarantee that there is a pit in (2,1) .in all cases KB true and alpha true.
So KB entails alpha.

Practice for knowledgebase $\neg P(1,1) \neg B(1,1)$ and Query: $P(2,1)$

